

Source:	C:\Users\isrys\OneDrive\Documents\KiCad\A1_Controller_Subsystem\A1_Sub\A1_Sub.kicad_sch																					
Date:	2026-03-07T16:57:17-0700																					
Tool:	Eeschema 9.0.7																					
Generator:	C:\Users\isrys\AppData\Local\Programs\KiCad\9.0\bin\scripting\plugins\bom_csv_grouped_by_value.py																					
Component Count:	70																					
Collated Components:																						
Item	Qty	Reference(s)	Value	LibPart	Footprint	Datasheet	DNP	Datasheet	Description	Footprint	Sim.Device	Sim.Pins	Cost per	Seller	Datasheet Link	Part #/Link	Receive	Surplus	Cost	Date Ordered		
1	26	3.3V_TP1, 8D_P~	~	IRS_Sub_Lib:TI	TestPoint:TestPoint_Loop_D2.50mm_Drill1.0mm				TestPoint:TestPoint_Loop_D2.50mm_Drill1.0mm				0	PCB		~	0	-26	\$0.00	03.07.2026		
2	9	B_P_DOWN1, B~	~	IRS_Sub_Lib:BT	All_footprints:SW_TL9320_EW1				Control Buttons	All_footprints:SW_TL9320_EW1			0.766	Digikey	https://configured	141-TL9320AF4	20	11	\$15.32	03.07.2026		
3	3	C1, C3, C4	100uF	Device:C_US	All_footprints:CAP_A768EB_6.8x6.8_KEM-M				capacitor, US syr	All_footprints:CAP_A768EB_6.8x6.8_KEM-M			0.327	Digikey	https://yagocgroup	399-A768EB107	10	7	\$3.27	03.07.2026		
4	1	C2	330uf	Device:C_US	All_footprints:WCAP-PSLP_6.3X7.7_DXL_				capacitor, US syr	All_footprints:WCAP-PSLP_6.3X7.7_DXL_			0.327	Digikey	https://www.we-o	732-6415-1-ND	3	2	\$0.98	03.07.2026		
5	1	D1	1N5819	Device:D	All_footprints:SC~		~		Diode	All_footprints:SCD		1=K 2=A	0.29	Digikey	https://www.diode	1N5819HW-FD1C	5	4	\$1.45	03.07.2026		
6	1	esp_32	~	IRS_Sub_Lib:ES	All_footprints:ESP32-S3-WROOM-1U				Brain	All_footprints:ESP32-S3-WROOM-1U			0	Prallta	https://www.espro	1965-ESP32-S3-WROOM-1-N4C		-1	\$0.00	03.07.2026		
7	1	F1	1A	IRS_Sub_Lib:Fu	All_footprints:696108003002				Fuse Holder	All_footprints:696108003002			0.62	Digikey	https://www.we-o	732-11376-ND	5	4	\$3.10	03.07.2026		
8	1	J1	USB_B_Micro	Connector:USB	Connector:USB:~		~		USB Micro Type Connector	USB:USB_Micro-B_Wuerth_614105150			0.78	Digikey	https://get.co/files	2073-USB83131-3	5	4	\$3.90	03.07.2026		
9	1	J2	Barrel_Jack_Swi	Connector:Barrel	Connector:Barre~		~		DC Barrel Jack v Connector	BarrelJack:BarrelJack_Horizontal			0	Prallta	https://www.cuide	CP-2519-ND	1	0	\$0.00	03.07.2026		
10	8	JP1, JP2, JP3, JP	Jumper_2_Small	Jumper:Jumper_	Jumper:jumper_~		~		Jumper, 2-pole, s	Jumper:jumper_0.8mm			0	PCB		~		-8	\$0.00	03.07.2026		
11	1	L1	220uH	Device:L	All_footprints:IN~		~		Inductor	All_footprints:IND_BOURNS_RLB9012			0	Prallta	https://www.bourn	RLB9012-221KL	1	0	\$0.00	03.07.2026		
12	1	LED_DEBUG_R~	~	IRS_Sub_Lib:LF	LED_SMD:LED_1206_3216Metric_Pad1.42x1.75mm_HandSolder				LED_SMD:LED_1206_3216Metric_Pad1.42x1.75mm				0.42	Digikey	https://mm.digike	160-1168-1-ND	3	2	\$1.26	03.07.2026		
	1	Above, but different led link for different color			green								0.39	Digikey	https://optoelectro	160-1456-1-ND	3	2	\$1.17	03.07.2026		
13	1	LM2575T-3.3G1	~	IRS_Sub_Lib:LA	All_footprints:D2PAK-5_ONS-M				V regulator	All_footprints:D2PAK-5_ONS-M			4.32	Digikey	https://www.onsej	LM2575D2T-3.3F	2	1	\$8.64	03.07.2026		
14	1	OLED1	~	IRS_Sub_Lib:OI	All_footprints:CONN4X1_NPTC041KFXC-RC_SUL				4_Pin Connect	All_footprints:CONN4X1_NPTC041KFXC-RC_SUL			0.57	Digikey	https://mm.digike	S5596-ND	5	4	\$2.85	03.07.2026		
15	1	P1	Conn_02x04_Do	IRS_Sub_Lib:8	All_footprints:CONN_332LF_AMP				8_Pin Connect	All_footprints:CONN_332LF_AMP			0.68	Digikey	https://cdn.amphic	609-5123-ND	1	0	\$0.68	03.07.2026		
16	1	P2	Conn_02x04_Up	IRS_Sub_Lib:8	All_footprints:CONN_332LF_AMP				8_Pin Connect	All_footprints:CONN_332LF_AMP			0.68	Digikey	https://cdn.amphic	609-5123-ND	1	0	\$0.68	03.07.2026		
17	9	R1, R4, R5, R6, 11k	~	Device:R_US	All_footprints:S1~		~		Resistor, US sym	All_footprints:STA_RMCF1206_STP-L			0.008	Digikey	https://www.seielc	RMCF1206FT1K	40	31	\$0.32	03.07.2026		
18	2	R2, R3	220	Device:R_US	All_footprints:S1~		~		Resistor, US sym	All_footprints:STA_RMCF1206_STP-L			0.01	Digikey	https://www.seielc	RMCF1206FT22	40	38	\$0.40	03.07.2026		
19	1	~	~						Fuse				0.436	Digikey	https://www.littelf	F5929-ND	5	4	\$2.18	03.07.2026		
20	1	OLED Screen							Screen				0	Prallta	https://www.amaz	I2C IIC SSD1306	1	0	\$0.00	03.07.2026		
																			Total Cost:	\$46.20		
																			Receipt link:	https://www.digikey.com/short/yfipm407		